# Waste Classification and Management Using Computer Vision

## Problem Description

### 1. Problem Definition:

The paper addresses the problem of **inefficient waste classification and management** in modern society. Municipal Solid Waste (MSW) consists of multiple categories such as plastic, paper, metal, glass, and organic waste, which must be properly segregated for recycling and disposal.

However, current systems rely heavily on **manual sorting**, which is:

- Time-consuming
- Costly
- Error-prone

As stated in the paper, only **about 10% of waste is recycled**, while the rest is improperly managed, leading to environmental pollution and health hazards

### 2. Importance of the Problem

The problem is highly significant because:

- Waste mismanagement leads to:
    - Air, water, and soil pollution
    - Global warming (methane emissions)
- Increasing urbanization is generating **huge amounts of waste daily**
- Efficient waste classification is essential for:
    - Recycling
    - Sustainable development
    - Smart city implementation

solving this problem directly impacts **environmental sustainability and public health**

## 3. Existing Solutions and Their Limitations

Existing approaches include traditional machine learning, CNN-based models, and IoT-based smart waste systems. However, these solutions suffer from low accuracy in complex environments, high computational cost, poor scalability, and limited real-time deployment capability. Some systems also fail to detect small objects, handle multiple waste categories, and adapt to region-specific waste types.

The paper reviews existing approaches such as:

- Traditional Machine Learning
- CNN-based models
- IoT-based smart waste systems

**Limitations:**

- Low accuracy in complex environments
- High computational cost
- Poor scalability
- Lack of real-time deployment capability

Some systems also fail to:

- Detect small objects
- Handle multiple waste categories
- Adapt to region-specific waste types

## 4. Challenges in the Problem

Waste classification is challenging due to visual complexity, as waste items vary in shape, size, and color. Different materials often appear similar, such as plastic and glass. Data variability across regions and class imbalance further complicate the problem. Additionally, real-world conditions like dirt, occlusion, and overlapping objects make accurate classification difficult.

## 5. Research Gap

There is a need for high accuracy with low computational cost. Existing models lack efficiency for real-time systems and often require high-end hardware. There is also limited use of region-specific training data. This paper addresses these gaps using EfficientNet-B0, transfer learning, and region-specific fine-tuning.

## 6. Objective of the Proposed Work

The main objective is to develop an efficient and accurate waste classification system using deep learning (EfficientNet-B0). The model leverages transfer learning and fine-tuning on specific datasets to improve performance. The final goal is to enable automated waste segregation for smart waste management systems.

## Research Paper Links -

**https://drive.google.com/drive/folders/172y_4591kUCp5N2ocohakAKSTZLNK4aX?usp=sharing**